

Engineering project dossier

Sofiene Zayati

A structured evidence record for CV tailoring, interview preparation, and professional-profile updates. It separates full-stack implementations, professional contributions, team projects, and frontend prototypes.

17

documented projects

93

portfolio screens

4

professional placements

4

working languages

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HOW TO USE THIS DOCUMENT

CV and interview guidance

For CV updates

Pair this dossier with the current CV. Select only the projects and technologies relevant to the target role, then rewrite bullets around contribution, scope, and verified implementation.

Claim discipline

Do not describe a frontend prototype as a production SaaS. Preserve labels such as team project, owned microservice, personal MVP, local-first product, and interactive prototype.

Best evidence

Prioritize Machetamache for full-stack product ownership, SmartProperty for team-scale AI integration, Prigado for professional automation, and MacroPark for connected embedded systems.

PROFESSIONAL POSITIONING

Full-Stack Web Engineer & Embedded Systems Graduate

I build full-stack platforms, AI-powered workflows, and connected products—combining modern web engineering with a practical embedded-systems foundation.

Frontend & Product

React · Next.js · Angular · TypeScript · JavaScript · Tailwind CSS

Backend & APIs

NestJS · FastAPI · Spring Boot · Laravel · Node.js · REST APIs · WebSockets

Embedded & IoT

Embedded C/C++ · ESP32 · WT32 · Arduino · MQTT · BLE · IoT Sensors

AI & Automation

n8n · Gemini · LLM Integration · PyTorch · XGBoost · Computer Vision · OCR

Data & Delivery

MongoDB · MySQL · Firebase · Docker · Kubernetes · Jenkins · CI/CD

Engineering Tools

Git · GitHub · Linux · Vercel · Cloudflare · SonarQube

PROJECT INDEX

Scope at a glance

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01

PERSONAL PRODUCT FOR A WORKING CHEF • 2026

Machetamache

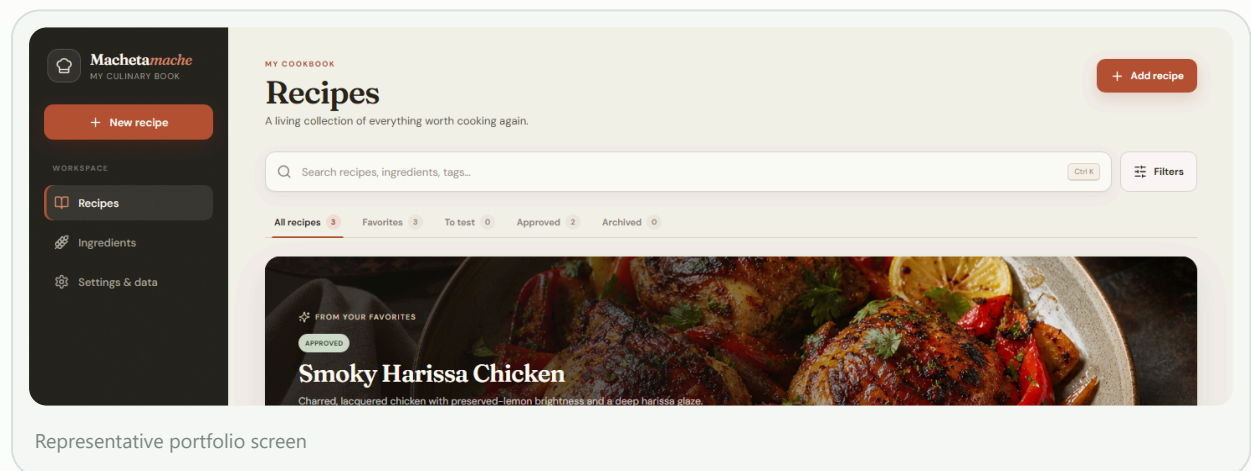
Local-First Recipe & Food-Costing Studio

Machetamache is a complete local-first product built for a real chef. It centralizes recipes, ingredient prices, portion scaling, production notes, profitability, photos, printing, and backups without requiring a cloud subscription. I engineered the React interface, Express API, SQLite data model, network-safety controls, automated tests, legacy-phone fallbacks, and branded Windows packaging from end to end.

CONTRIBUTION Product designer & full-stack engineer

DECLARED SCOPE Full-stack local product • Windows release

PRIMARY CATEGORY web + mobile



Representative portfolio screen

TECHNOLOGY EVIDENCE

React 19

TypeScript

Express 5

SQLite

Vitest

Supertest

Vite

PowerShell

IMPLEMENTED CAPABILITIES

Recipe Library

Photos, search, filters, categories, favorites, archive, duplication, ordered steps, and production notes.

Portion Scaling

Non-destructive quantity scaling with the option to preserve the result as a recipe variation.

Exact Food Costing

Ingredient price normalization, extra expenses, per-portion cost, revenue, profit, and margin calculations.

Sub-Recipes & Versions

Reusable preparations and version history support practical kitchen iteration.

Print-Ready Sheets

Kitchen and price-free print modes produce useful handoff documents instead of browser screenshots.

Private Wi-Fi Access

The same cookbook opens on the Windows host and phones connected to its private network.

Backup & Restore

Full JSON export and restore protects the cookbook independently from the application package.

Portable Windows Package

A branded launcher, bundled runtime, visible control panel, and external app-data storage make deployment approachable.

ARCHITECTURE AND ENGINEERING

One Cookbook, Two Device Classes

A React client on the Windows PC and private-network phones talks to one Express server. The server owns the SQLite database, static production bundle, security checks, and controlled shutdown behavior.

Practical Local-First Distribution

The launcher discovers the current private IP, starts the server, opens the local browser, and shows the phone URL. Modern and legacy bundles, UUID fallback behavior, and an early preflight screen cover real private-HTTP phone constraints.

PROBLEMS SOLVED

Money Without Rounding Drift: Food-cost and margin calculations use integer milli-euros rather than floating-point currency values.

Private HTTP on Real Phones: Fallback UUID generation, a legacy bundle, compatible content security rules, and early error reporting address browser differences outside localhost.

Desktop-Grade Delivery from a Web Stack: The product needed one-click Windows startup, visible server control, persistent data, and safe shutdown without asking the chef to use a terminal.

OUTCOME AND CV-READY EVIDENCE

One Shared Cookbook

PC and phone clients work against the same local data without cloud hosting or recurring infrastructure.

Verified Full Stack

Type checks, unit tests, API tests, production builds, and release packaging are part of the repository workflow.

Built Around Kitchen Work

Scaling, costing, printing, backups, and touch layouts solve daily operational tasks rather than presenting a static recipe catalog.

Distributable Product

A portable Windows release packages the server, runtime, launchers, control panel, and user guidance.

Repository [↗](#)

SmartProperty

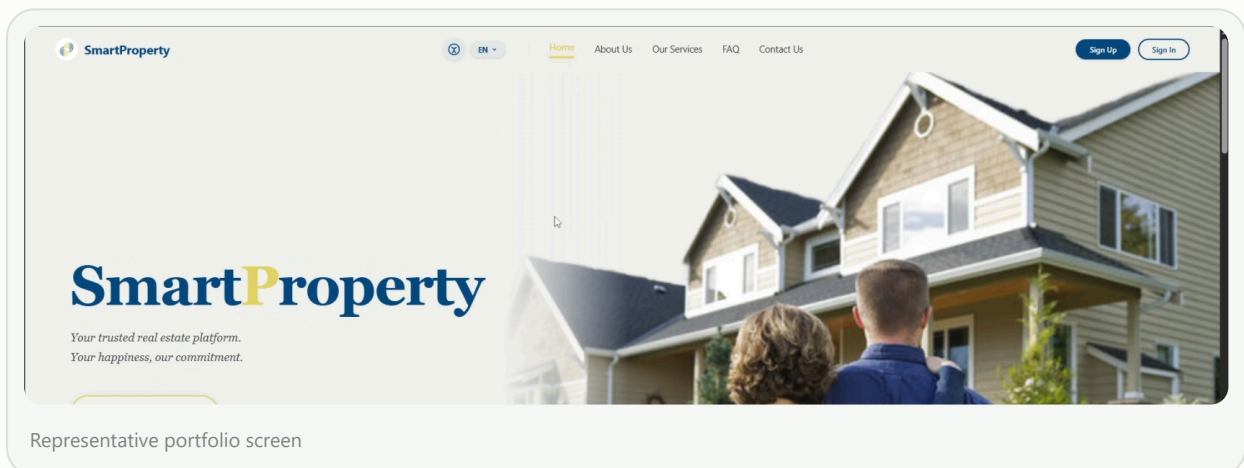
AI-Powered Real Estate SaaS

SmartProperty is a multi-role real estate platform for agencies, owners, accountants, and clients. I contributed to application workflows, interface and notification improvements, solvency-analysis integration, testing, and cross-branch integration within a larger team system spanning web services, machine learning, OCR, and computer vision.

CONTRIBUTION Full-stack & AI contributor

DECLARED SCOPE Team-built product prototype

PRIMARY CATEGORY web + ai



TECHNOLOGY EVIDENCE

React

NestJS

FastAPI

MongoDB

XGBoost

Socket.IO

Docker

Kubernetes

Jenkins

IMPLEMENTED CAPABILITIES

Property Listings Management

Centralized listings with images, documents, location, and lifecycle status — managed by agents, owners, and admins across multiple branches.

Property Recommendation Engine

A recommendation feed ranks listings using budget, preferences, property data, and location criteria.

Application Review Signals

Affordability, occupation, and document signals give reviewers structured context for an applicant decision.

AI Property Valuation

XGBoost valuation model trained on historical data to predict optimal property prices and rental rates with market trend context.

Predictive Pricing

Automatic rent estimation from amenities, location, and property type — filling in the price field on listings with one click.

AI Image Description Generation

CLIP and BLIP analyze uploaded property photos and generate room-by-room listing copy, accelerating marketing content production.

OCR Document Analysis

OCR.space extracts structured data from income, identity, and supporting documents to feed the solvency pipeline.

Fraud Detection

Income mismatch detection and cross-field validation flag suspicious applications before they reach the approval queue.

Solvency Analysis

Prototype risk scoring that combines affordability ratios, occupation validation, and document evidence for review.

AI Chat Assistant

Groq-powered conversational agent that estimates budgets, answers property questions, and guides users through the platform.

Voice-Enabled Navigation

Hands-free search and navigation that listens to natural-language queries and routes users to the right view or listing.

Multi-Role Dashboards

Tailored experiences for real-estate agents, admins, clients, and accountants — each with KPIs and quick actions specific to the role.

Engagement Analytics

Per-property views, clicks, saves, and unique users tracked at the branch level to measure listing performance.

Visit Request Management

Multi-slot visit scheduling with date and time pickers, application status, and visit history for clients.

Property Management Workflows

Add-property flow with map-based location picking, image uploads, AI price estimation, and publish-to-market steps.

Real-Time Updates

Socket.IO notifications, dashboard refresh, and event broadcasting across roles for immediate operational feedback.

ARCHITECTURE AND ENGINEERING

Microservices Architecture

Independent services communicate over REST and Socket.IO, separating product workflows from ML and computer-vision capabilities.

AI & OCR Pipeline

Specialized models and OCR services feeding recommendations, valuation, and solvency flows.

DevOps & Quality

The repositories include container, CI/CD, quality-gate, and Kubernetes deployment configuration; production deployment is outside the case-study claim.

PROBLEMS SOLVED

Multi-Role Authorization: Four distinct user roles (agent, admin, client, accountant) with overlapping yet isolated permissions, branches, and dashboards.

Real-Time Events: Socket.IO event broadcasting across roles for dashboard refresh, notifications, and engagement metrics without overwhelming the client.

OCR + AI Pipeline Reliability: Combining OCR.space extraction, XGBoost scoring, and rule-based validation into a single, auditable solvency analysis.

Computer Vision Latency: CLIP and BLIP inference required careful user feedback and service boundaries so image-description generation remained understandable in the interface.

Accessibility vs Density: Balancing a feature-rich dashboard with keyboard-friendly navigation, voice input, readable structure, and accessible interaction patterns.

OUTCOME AND CV-READY EVIDENCE

Valuation Model

A prototype pricing model trained on historical Tunisian listings to support—not replace—human valuation.

Property Ranking

Recommendation flows surface relevant properties using preferences, affordability, and listing data.

Operations

OCR extraction and workflow automation reduce repetitive steps across administrative roles.

Solvency Review

Document extraction and rule-based signals create an auditable starting point for manual review.

Accessible Use

Keyboard-aware flows, voice input, and image descriptions broaden how users can navigate the product.

Demo [↗](#)

03

CLIENT-READY PRODUCT CONCEPT · 2026

Aurelle

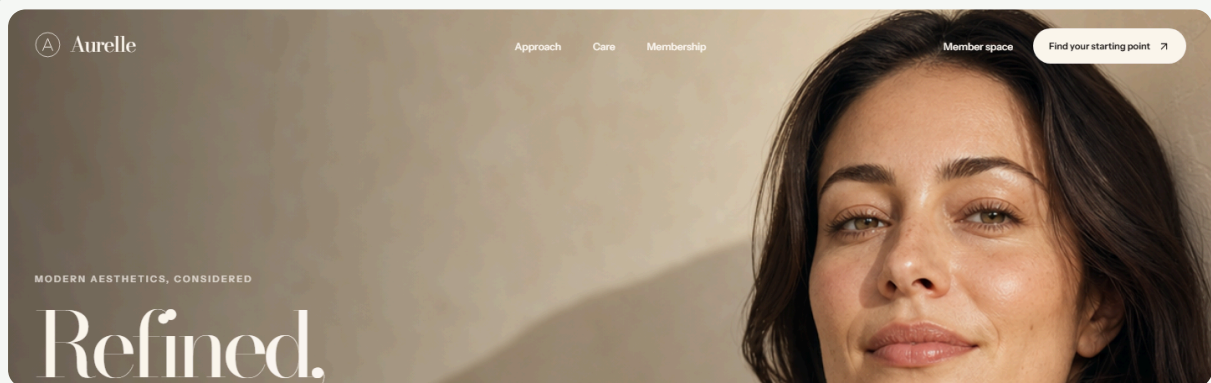
Premium Aesthetics Client Experience

Aurelle is a portfolio-grade React product concept for a premium aesthetics clinic. I designed and built the full responsive experience around two business goals: converting mobile visitors into consultation requests and supporting retention through an organized member space. It includes guided goal discovery, appointment selection, browser persistence, accessible motion, medical disclaimers, and automated accessibility and visual smoke checks.

CONTRIBUTION **Product designer & frontend engineer**

DECLARED SCOPE **Interactive frontend · Automated QA**

PRIMARY CATEGORY **web + mobile**



Representative portfolio screen

TECHNOLOGY EVIDENCE

React 19

React Router

JavaScript

Motion

Playwright

axe-core

Vite

IMPLEMENTED CAPABILITIES

Editorial Landing Experience

A premium visual system communicates trust, restraint, treatment philosophy, and clear next actions.

Guided Consultation

Goal-based questions lead to contextual recommendations while maintaining clear non-medical boundaries.

Demo Booking Flow

Treatment context, date and time selection, confirmation, and persistence create a complete conversion journey.

Member Care Portal

Upcoming care, credits, routine, timeline, preparation notes, billing, and concierge messaging live in one space.

Accessible Interaction

Semantic controls, focus states, mobile navigation, reduced motion, and automated checks support inclusive use.

Browser Smoke Journeys

Automated desktop and mobile flows cover discovery, consultation, booking, portal navigation, and overflow checks.

ARCHITECTURE AND ENGINEERING

Connected Frontend Product Story

The public experience, guided consultation, booking drawer, and member route share a small browser persistence layer. This makes the concept testable as one client journey while keeping production API, authentication, scheduling, and payment boundaries explicit.

PROBLEMS SOLVED

Premium Without Friction: The visual language needed to feel editorial while keeping navigation, calls to action, and text legible.

Safe Medical UX: The consultation experience provides useful orientation without presenting a frontend concept as clinical advice.

One Continuous Journey: A demo appointment selected on the public site must remain visible and meaningful inside the member portal.

OUTCOME AND CV-READY EVIDENCE

Three Connected Journeys

Discovery, consultation and booking, and ongoing member care form one coherent product narrative.

Repeatable Browser Verification

Visual smoke checks exercise the key desktop and mobile journeys instead of relying on static screenshots alone.

Accessibility Included

Automated axe checks and interaction-level accessibility decisions are built into the prototype workflow.

Pitchly

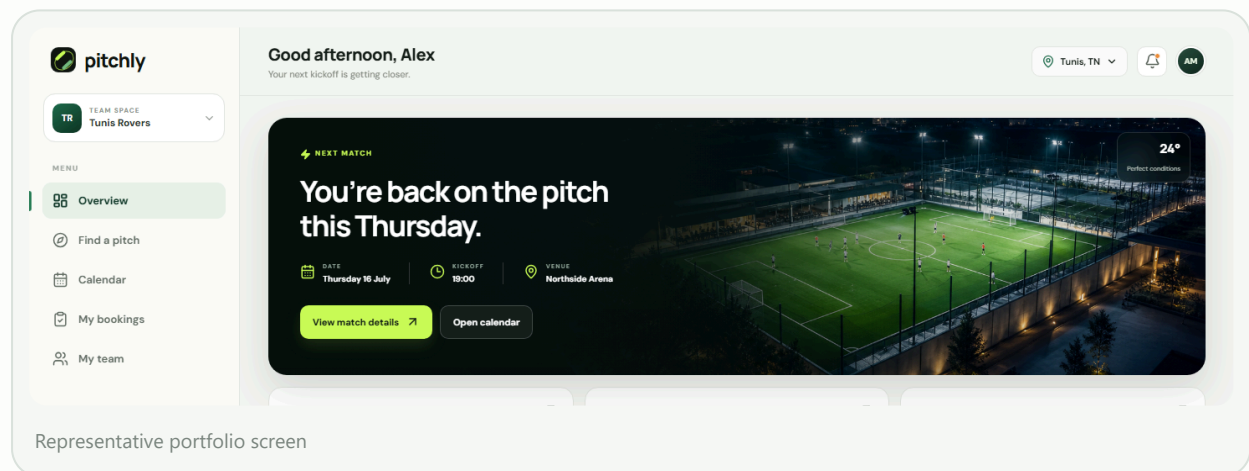
Two-Sided Football Pitch Marketplace

Pitchly is a responsive two-sided product prototype for football-pitch reservations. Players can discover venues, choose slots, manage bookings, and coordinate a team; venue managers can review the same requests, approve or decline them, manage schedules, inspect revenue signals, and maintain venue information. Shared browser data makes both roles feel like one coherent system while keeping the current frontend-only scope explicit.

CONTRIBUTION Product designer & frontend engineer

DECLARED SCOPE Responsive frontend • Shared demo state

PRIMARY CATEGORY web + mobile



Representative portfolio screen

TECHNOLOGY EVIDENCE

React 19

TypeScript

Vite

Lucide

LocalStorage

Responsive CSS

IMPLEMENTED CAPABILITIES

Venue Discovery

Search, filters, favorites, list and map views, amenities, formats, ratings, distance, and pricing.

Reservation Workflow

Date, time, duration, note, price, confirmation, cancellation, and status-aware booking details.

Team Space

Roster, captain status, join code, invitations, next match, and team performance context.

Calendar & Notifications

Month and week schedule views connect booking changes to visible player and manager follow-up.

Approval Queue

Venue staff review the same pending requests and can approve or decline them from focused operational views.

Venue Operations

KPIs, revenue signals, occupancy, schedule, venue settings, hours, and availability support the supply side.

ARCHITECTURE AND ENGINEERING

One Prototype, Two Operational Perspectives

A shared booking domain powers player and manager presentations. URL parameters make either role directly reviewable, while local persistence lets approvals and cancellations carry between the two views during a demo.

PROBLEMS SOLVED

Two Sides of One Booking: Player and manager views needed to read as different tools while remaining consistent about status and schedule.

Operational Density on Mobile: Calendars, status details, teams, and approval actions required responsive information prioritization.

Prototype Honesty: The interface simulates shared state but clearly identifies authentication, payments, concurrency, and server-side slot locking as backend work.

OUTCOME AND CV-READY EVIDENCE

Complete Two-Sided Narrative

Both marketplace demand and supply workflows can be demonstrated from the same frontend.

Five Booking States

Pending, confirmed, declined, completed, and cancelled states shape the relevant actions and messaging.

Responsive Product System

Desktop, tablet, and mobile layouts support both role families without reducing the demo to a dashboard image.

Repository [↗](#)

Interna

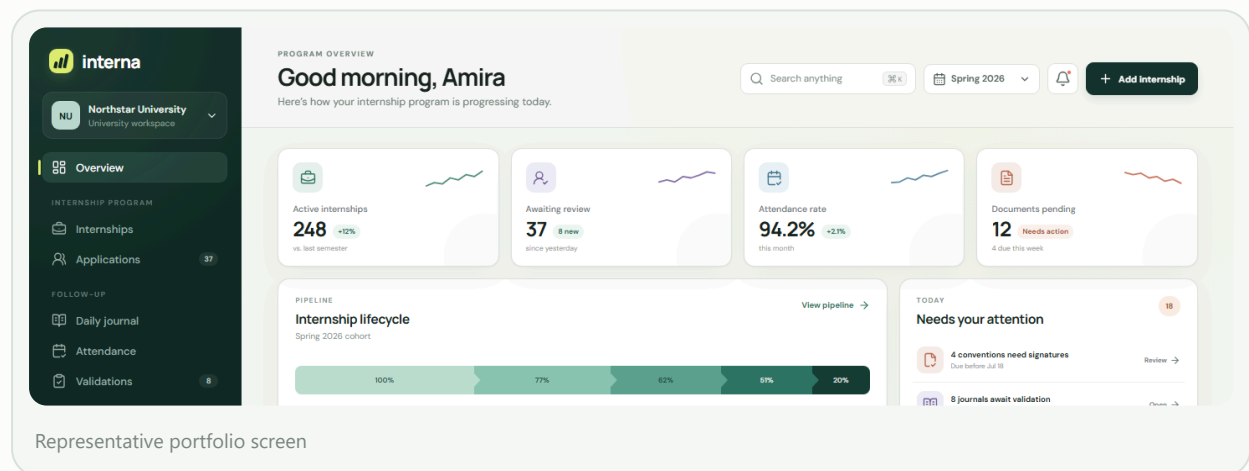
University Internship Operations Platform

Interna translates the full internship lifecycle into a responsive administration and student experience. University teams can follow cohorts, placements, approval pipelines, attendance exceptions, documents, validations, and evaluations, while students see their own progress, deadlines, journal, and report readiness. The current version is an API-ready frontend prototype backed by realistic local demo data.

CONTRIBUTION Product designer & frontend engineer

DECLARED SCOPE Dual-role frontend · API-ready

PRIMARY CATEGORY web + ai, mobile



TECHNOLOGY EVIDENCE

React 19

JavaScript

Vite

Lucide

Responsive CSS

Local demo data

IMPLEMENTED CAPABILITIES

Cohort Overview

Lifecycle pipeline, active placements, approval rates, attendance, deadlines, and an action center.

Placement & Application Pipeline

Students, companies, supervisors, dates, progress, stages, and attention states in connected views.

Journals & Validations

Daily entries, review queues, milestones, supervisor feedback, and pending validation work.

Attendance Operations

Program-level rate, exceptions, late arrivals, absences, and justification follow-up.

Documents & Evaluations

Conventions, certificates, signatures, due dates, grading states, and final assessment.

AI-Assisted Reports

Report readiness, draft organization, progress inputs, and a dedicated assisted-writing workspace.

Personal Internship Workspace

Individual progress, next steps, journal, attendance, deadlines, documents, and report readiness.

ARCHITECTURE AND ENGINEERING

Dual Information Architecture

One React application switches between the university program office and the individual student perspective. Navigation, page metadata, dashboards, and task emphasis change by role while sharing a coherent internship domain.

PROBLEMS SOLVED

A Long Lifecycle, One Mental Model: Applications, conventions, attendance, journals, validations, documents, and grades needed to feel connected rather than like separate tools.

Role-Appropriate Density: Program administrators need portfolio-level oversight while students need a calm personal task view.

Useful AI Placement: Assistance is positioned inside report readiness and validated journal evidence rather than as an isolated chatbot.

OUTCOME AND CV-READY EVIDENCE

Two Complete Workspaces

A program administrator and a student can each demonstrate a role-specific end-to-end experience.

Eight Lifecycle Domains

Placements, applications, journals, attendance, validations, documents, reports, and evaluations share one system.

Clear Backend Boundary

The current frontend communicates a realistic product model while remaining explicit about the API work required for production.

Repository [↗](#)

Prigado AI

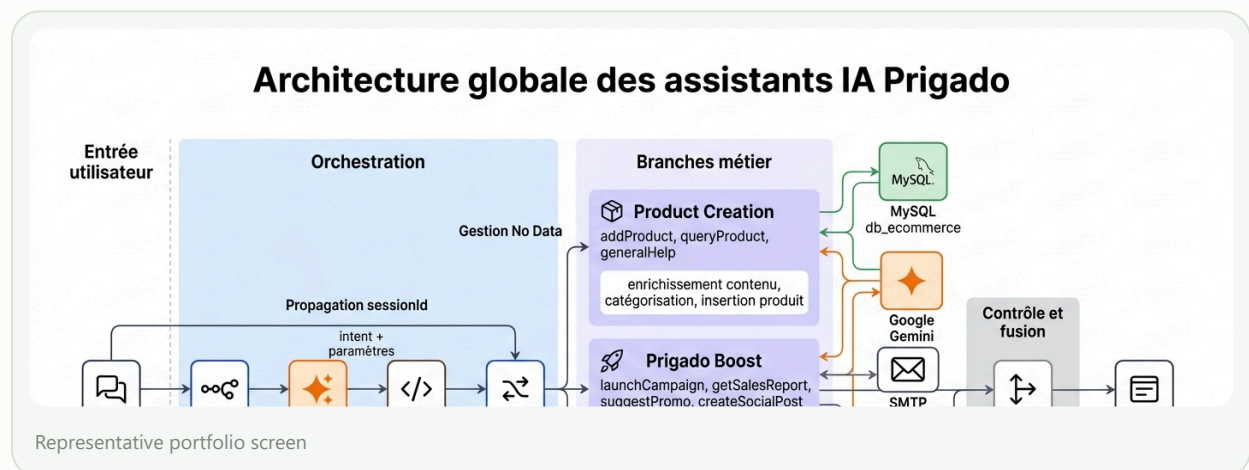
Intelligent E-Commerce Automation

During my Inspark Connect placement, I developed AI-powered workflows and assistants for the Prigado e-commerce platform. The work included n8n routing, intent extraction, JSON sanitization, structured SQL aggregation, Gemini integrations, and automated email flows connected to a Laravel and Vue.js product.

CONTRIBUTION AI automation & full-stack developer

DECLARED SCOPE Integrated into an existing platform

PRIMARY CATEGORY ai + web



TECHNOLOGY EVIDENCE

n8n

Laravel

Vue.js

MySQL

Gemini

SMTP

REST APIs

IMPLEMENTED CAPABILITIES

Product Creation Assistant

Multi-path workflow handling add product, query product, and general help intents with price enrichment, category suggestion, tag generation, and MySQL insertion.

Boost — Marketing & Sales

Automated email campaigns, sales reports by seller, bundle suggestions, and social post generation based on business data.

Logistics Intelligence

Peak activity prediction, trending product ranking, and customer risk classification using order history and delivery patterns.

AI Router & Intent Extraction

Strict intent routing with JSON guardrails, sanitized AI outputs, and consistent payload structures across all workflow branches.

SQL Aggregation for AI

Aggregate nodes consolidate relational SQL rows into structured arrays, ensuring AI agents receive clean, readable inputs for accurate responses.

Business-Ready Outputs

Reports, recommendations, and campaign content delivered in formats ready for decision-makers without manual processing.

ARCHITECTURE AND ENGINEERING

System Architecture

AI assistants sit on top of the Prigado Laravel+Vue.js stack, orchestrating MySQL, Gemini, and external APIs through n8n workflows.

Multi-Path Routing

Intent-driven routing directs user requests to specialized workflow paths based on extracted intent and parameters.

Use Case Coverage

Comprehensive coverage across admin, seller, and customer workflows.

PROBLEMS SOLVED

Guard Clauses for Empty Data: Explicit fallback responses for missing or empty SQL results to prevent ambiguous AI answers.

Consistent Payloads Across Branches: Set and Code nodes normalize JSON structures before merges to keep outputs stable across all paths.

Sanitized AI Outputs: Regex-based cleanup ensures strict JSON compliance for downstream nodes and reduces runtime parsing errors.

Relational Joins for Analytics: Precise joins between users, orders, order_details, products, and categories to deliver accurate business insights.

OUTCOME AND CV-READY EVIDENCE

Efficiency

Automated key business processes across product creation, sales, marketing, and logistics — reducing manual effort significantly.

Data Quality

Standardized SQL aggregations and sanitized AI outputs ensure consistent, reliable data for decision-making.

Speed

Delivered fast, readable outputs ready for operators and executives without additional processing.

TounsiAds

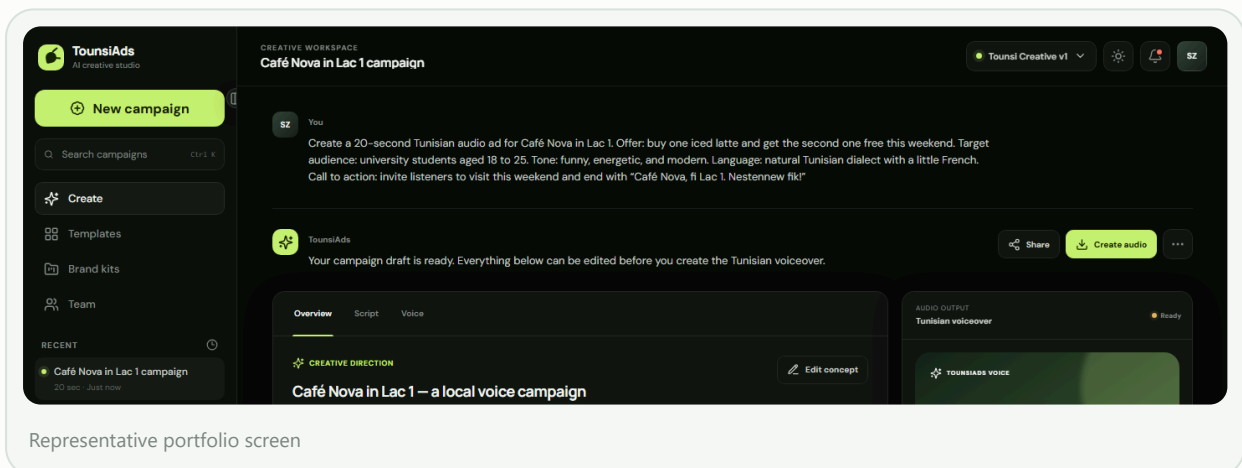
Tunisian-Dialect Campaign & Voice Studio

TounsiAds is a personal full-stack MVP for building local advertising campaigns. Users can draft and edit concepts and scenes, manage brand and template settings, use browser speech input, and generate or download WAV voiceovers. The Gemini key remains on an Express server route, while drafts and workspace preferences are persisted locally for the prototype.

CONTRIBUTION Product designer & full-stack developer

DECLARED SCOPE Working MVP • Local persistence

PRIMARY CATEGORY ai + web



Representative portfolio screen

TECHNOLOGY EVIDENCE

React

Vite

Node.js

Express

Gemini TTS

Web Speech API

WAV

IMPLEMENTED CAPABILITIES

Dynamic Drafts

Business, offer, audience, tone, duration, and call-to-action inputs create an editable campaign structure.

Editable Workspace

Concept, audience, tone, individual scenes, and final transcript remain editable after generation.

Templates & Brand Kit

Starter prompts, reusable brand details, team prototype, settings, help, and campaign history in one workspace.

Speech Input

Browser speech-to-text can capture a brief in supported browsers, alongside typed prompts and attachments.

Tunisian Voiceover

Selectable Gemini voices and delivery directions generate playable audio through a protected server route.

WAV Download

Raw model audio is converted into a standard WAV file for native playback and export.

ARCHITECTURE AND ENGINEERING

Browser Workspace + Server-Side TTS

React manages campaign editing and local history. Vite proxies audio requests to an Express API, where the Gemini credential remains server-side. The response is converted to WAV and returned for browser playback or download.

PROBLEMS SOLVED

Credential Boundary: Keeping the model key out of the React bundle while preserving a simple local full-stack development flow.

Editable AI Output: Treating generated content as a draft that users can inspect and refine, rather than an opaque final result.

Dialect Variability: Supporting Arabic, Tunisian Latin/Arabizi, and mixed French wording while making pronunciation limitations visible.

OUTCOME AND CV-READY EVIDENCE

Brief to Playable Audio

One connected MVP flow turns a local campaign idea into editable copy and a downloadable voiceover.

Safer AI Integration

Provider credentials stay on the Node server rather than being exposed through Vite environment variables.

Human-in-the-Loop Workflow

Every generated concept, scene, and transcript remains editable before voice generation.

MacroPark

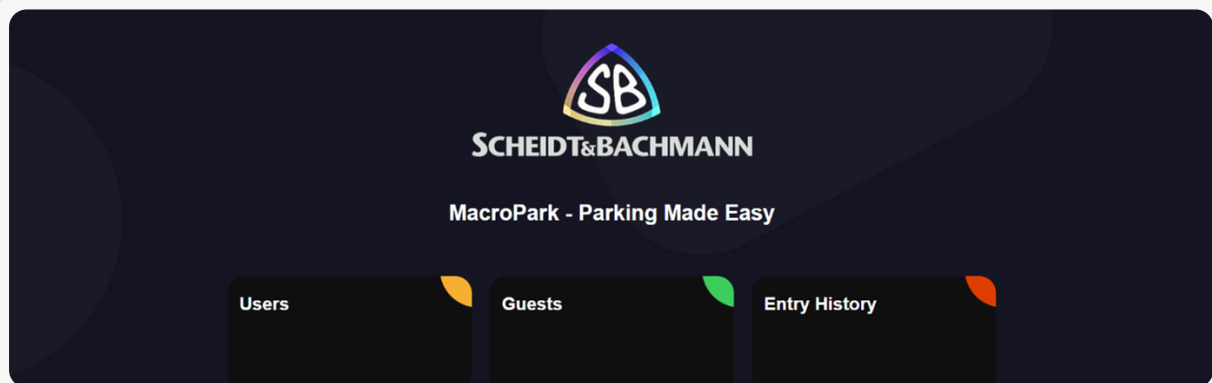
Connected Parking Access & Supervision System

MacroPark was my 2024 graduation project at Scheidt & Bachmann Maghreb, co-developed in a two-person engineering team across five Scrum sprints. It delivers the complete parking workflow: administrators manage users, guests, access evidence, barriers, emergency commands, and event logs; drivers enter through camera-based plate recognition or the Flutter app over BLE; FastAPI, Core, Bridge, MQTT, and database services coordinate decisions; and WT32-ETH01 firmware controls four barrier channels with persistent configuration, Ethernet/Wi-Fi connectivity, and OTA maintenance.

CONTRIBUTION Co-developer in a two-person engineering team

DECLARED SCOPE Complete multi-application IoT system

PRIMARY CATEGORY embedded + mobile, web



Representative portfolio screen

TECHNOLOGY EVIDENCE

C++ / Arduino

WT32-ETH01

Flutter

FastAPI

MQTT

BLE

Quercus SmartLPR

Docker Compose

IMPLEMENTED CAPABILITIES

Administration Platform

A central dashboard connects user, guest, entry-history, barrier, emergency-control, and barrier-event modules.

Users & Guest Access

Administrators manage registered employees, license plates, invited visitors, approvals, and access credentials.

Traceable Entry History

Every access record can include user, method, device, time, recognized plate, and captured LPR evidence.

Barrier Operations

Installed barriers, individual command channels, emergency actions, and controller event logs are managed from the web platform.

Camera-Based Entry

A Quercus SmartLPR camera captures plates, passes them through Bridge and Core services, and triggers an approved barrier command.

Flutter BLE Entry

Authenticated drivers can request barrier access from the Android app, then review their profile and camera or phone access history.

Four-Barrier Firmware

The WT32 firmware drives four independent open/close relay pairs and supports timed automatic closing for mobile commands.

Provisioning & Persistence

A local configuration portal manages IP and MQTT settings; SPIFFS preserves network and company-ID configuration across restarts.

OTA Maintenance

Arduino OTA handlers and the controller web portal provide a practical remote firmware-update workflow.

Containerized Service Stack

Docker Compose starts the Backend, Core, LPR/Bridge, and Mosquitto services together with explicit ports and topics.

ARCHITECTURE AND ENGINEERING

Two Access Paths, One Parking System

MacroPark is the complete product. Its architecture map shows how the admin platform, mobile app, Quercus camera, Bridge server, Core server, Backend server, databases, Mosquitto broker, WT32 controller, and physical barrier work together.

WT32 Edge-Control Boundary

The WT32-ETH01 connects network communication to physical relay outputs. The supplied 566-line firmware covers Ethernet and Wi-Fi setup, MQTT subscriptions, BLE scanning, four barrier channels, timed closing, SPIFFS configuration, reset behavior, and OTA updates.

PROBLEMS SOLVED

Coordinating a Distributed Product: The working system required three application servers, an MQTT broker, databases, a camera, a mobile client, and an embedded controller to agree on access decisions and logs.

Two Access Paths, Consistent Control: Camera/MQTT and phone/BLE entry had to converge on the same physical barrier behavior while preserving the source and evidence for each event.

Reliable Edge Configuration: The controller needed to remain configurable when infrastructure changed, persist critical network and company settings, and expose a reset path.

From Software to Physical Validation: Relay wiring, four control channels, timed closing, permissions, BLE payloads, MQTT topics, and actual barrier responses were tested as one system.

OUTCOME AND CV-READY EVIDENCE

Four Integrated Product Layers Admin web, Flutter mobile, LPR services, and embedded barrier control form one complete parking product.
Two Verified Entry Methods The presentation and report document successful camera/MQTT and mobile/BLE barrier-opening paths.
Five-Sprint Delivery Implementation progressed through platform, mobile, controller, LPR, and test-and-validation sprints.
Hardware-Backed Prototype Controller configuration, relays, BLE events, MQTT commands, OTA behavior, and access history were validated beyond static interface mockups.

Firmware [↗](#)

Genius PMO HRMS

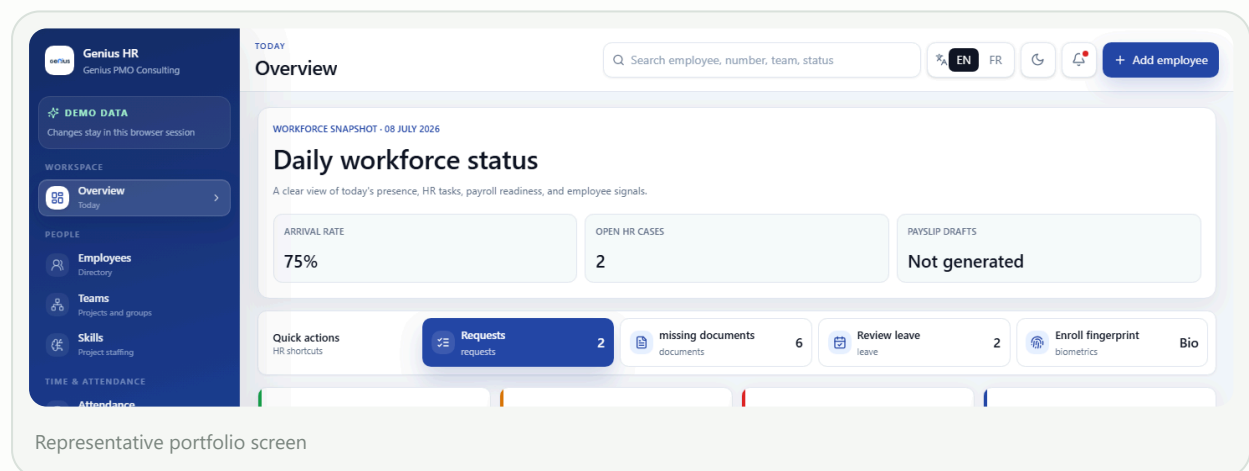
HR Operations Dashboard Prototype

Genius PMO HRMS is a high-fidelity HR product prototype built around realistic day-to-day operations. The updated frontend provides four role-specific demo workspaces, responsive English and French modes, light and dark themes, accessible charts, CSV export, draft payslip generation, and carefully labeled interactions. A FastAPI, SQLAlchemy, Alembic, and PostgreSQL backend foundation supplies health and database plumbing while authentication and HR business endpoints remain explicit future work.

CONTRIBUTION Product & frontend engineer

DECLARED SCOPE Interactive frontend · Backend foundation

PRIMARY CATEGORY web



Representative portfolio screen

TECHNOLOGY EVIDENCE

Next.js 15

React 19

TypeScript

Tailwind CSS

FastAPI

PostgreSQL

PDF generation

CSV export

IMPLEMENTED CAPABILITIES

HR Overview

A compact daily cockpit for workforce presence, payroll readiness, document gaps, leave, requests, and exception queues.

Attendance Review

Schedule variance, late or missing scans, source filters, exceptions, and an HR review workflow.

Employee Records

Searchable employee directory and detailed records covering assignment, schedule, documents, payroll, skills, and history.

Teams & Project Staffing

Team structure, capacity, project assignments, role coverage, and skills help connect people records to active delivery work.

Document Health

Required-document tracking, employee checklists, reminders, and clearly simulated upload or status actions.

Draft Payroll

Period preparation with attendance adjustments, readiness checks, previews, and individual or batch draft payslips.

HR Case Center

Requests organized by priority, owner, SLA, activity timeline, response draft, status, and resolution.

Reports & Exports

Accessible workforce charts with working UTF-8 CSV exports and operational summaries.

Four Demo Workspaces

Dedicated HR, manager, employee, and executive entry points demonstrate how the same operations adapt to different responsibilities.

Bilingual & Responsive

Persisted English/French and light/dark preferences across desktop and mobile dashboard layouts.

ARCHITECTURE AND ENGINEERING

Interactive Frontend + Backend Foundation

Next.js routes and shared dashboard components use a typed client-side provider and explicitly labeled mock HR data. A FastAPI, SQLAlchemy, Alembic, PostgreSQL, and Docker foundation supplies health and database-readiness boundaries; authentication and production HR endpoints remain future work.

PROBLEMS SOLVED

Dense Workflow Clarity: Organizing many HR modules without turning the product into a collection of oversized, disconnected dashboard cards.

Honest Prototype States: Making biometric, payroll, upload, and notification simulations useful while clearly distinguishing them from connected production behavior.

Operational Responsiveness: Keeping data-heavy tables, drawers, filters, and navigation usable on smaller screens.

OUTCOME AND CV-READY EVIDENCE

Coherent HR Workspace

A navigable prototype that demonstrates role-aware journeys across daily workforce operations.

Explicit Integration Boundary

Typed domain models and a working backend foundation make the next production integration step clear.

Credible Demo Scope

Working interactions, exports, themes, and language preferences without disguising mock data as production infrastructure.

Green Coffee OS

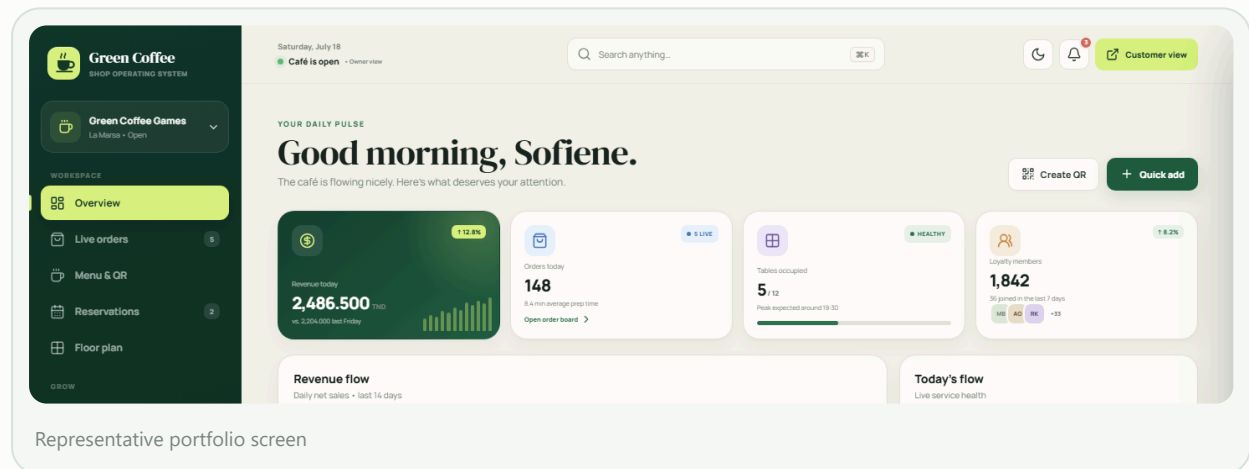
Coffee Shop Operations SaaS Prototype

Green Coffee OS translates a broad coffee-shop feature set into a coherent, responsive product experience. The prototype supports owner, manager, barista, floor-staff, and customer demo journeys with shared local data, protected demo routes, light and dark themes, and detailed operational modules. Production authentication, payments, messaging, realtime sync, and AI integrations remain clear next-step backend work.

CONTRIBUTION Product designer & frontend engineer

DECLARED SCOPE Frontend prototype · Browser persistence

PRIMARY CATEGORY web



TECHNOLOGY EVIDENCE

React 19

Vite

JavaScript

CSS

LocalStorage

SessionStorage

IMPLEMENTED CAPABILITIES

Live Order Board

QR and staff-entered orders move through new, preparing, ready, and served states with barista and KDS views.

Floor & Reservations

Table sessions, occupancy, secure table QR flows, reservations, calendar, waitlist, guest messages, and event bookings.

Menu Management

Categories, stock, modifiers, allergens, availability, featured products, and customer menu preview.

CRM & Loyalty

Customer profiles, preferences, receipt history, rewards, VIP tiers, birthdays, and referrals.

Campaign Workflows

Segments, coupons, feedback QR, ratings, and prototype WhatsApp, email, and push campaign experiences.

Operational Analytics

Revenue, order, product, reservation, QR-scan, customer, and peak-hour views for decision support.

ARCHITECTURE AND ENGINEERING

Shared Interactive Frontend

React modules share one browser-persisted shop dataset so changing demo roles feels like working in the same café. Role guards and session state shape each workspace, while production services remain an explicit future boundary.

PROBLEMS SOLVED

Turning Scope into Product: Converting a large capability inventory into coherent navigation and workflows rather than a feature checklist.

Shared Multi-Role State: Making role switching feel connected while keeping the implementation honest as a browser-only prototype.

Operational Density: Balancing fast staff workflows with readable detail across desktop, tablet, and mobile layouts.

OUTCOME AND CV-READY EVIDENCE

Role-Focused Journeys

Five distinct demo experiences demonstrate how one product can serve the whole coffee-shop operation.

Connected Demo State

Orders, menu availability, tables, reservations, and automations persist across staff role changes.

Product-Ready UI Direction

A polished, responsive interface that provides a concrete foundation for backend discovery and implementation.

Repository [↗](#)

InnoMall

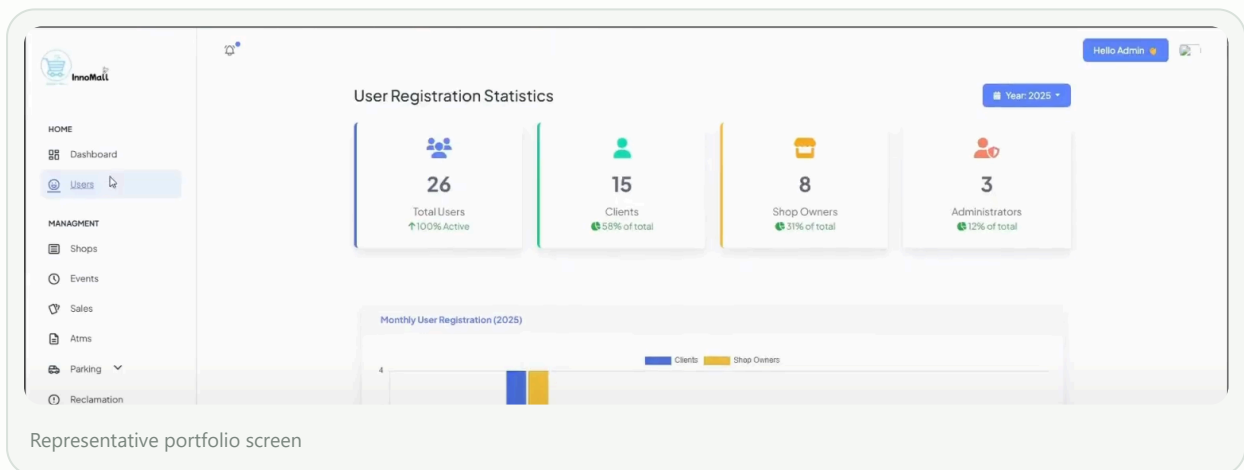
Integrated Mall Management Ecosystem

InnoMall combines a JavaFX administration application and a Symfony web platform for customers and shop owners. I contributed to a team system that connected multi-floor parking sensors, live availability, online reservations, Stripe payment flows, a Gemini assistant, Twilio notifications, and operational analytics.

CONTRIBUTION Full-stack & IoT contributor

DECLARED SCOPE Multi-application team prototype

PRIMARY CATEGORY web + embedded



TECHNOLOGY EVIDENCE

Symfony 6

Java

JavaFX

MySQL

Stripe API

Gemini API

Twilio

IoT Sensors

IMPLEMENTED CAPABILITIES

Dual Interface System

JavaFX desktop for administrators and Symfony web for customers, providing tailored experiences for different user types.

IoT Multi-Floor Parking

Real-time parking availability tracking using IoT sensors with multi-floor support, live spot monitoring, and online reservation system.

Integrated Payments

Secure payment processing through Stripe API supporting multiple payment methods and automated transaction management.

AI Customer Support

A Gemini-powered chatbot interface supports FAQs and store-information queries in the team prototype.

Analytics Dashboard

Comprehensive analytics on customer behavior, sales trends, operational efficiency, and parking usage patterns.

Multi-Channel Notifications

Automated Twilio SMS system for promotions, reservation confirmations, and important operational alerts.

ARCHITECTURE AND ENGINEERING

Backend Technologies

Symfony 6 provides the web backend with Doctrine ORM for database management and Twig templating for the web interface.

IoT & Hardware Integration

ESP32-based sensor network for real-time multi-floor parking management.

PROBLEMS SOLVED

Real-Time Data Flow: Connecting MQTT parking updates to customer and administrator interfaces while keeping occupancy state understandable.

Secure Payments: Stripe tokenization keeps raw payment-card data outside the application while supporting online checkout.

Cross-Platform Compatibility: Ensuring consistent behavior between JavaFX desktop and Symfony web interfaces with shared business logic.

IoT Integration: Coordinating parking sensors, reservation flows, live availability, and application state across the team system.

OUTCOME AND CV-READY EVIDENCE

Operational Flow

• Live parking-state presentation • Reservation and notification workflows • Payment and analytics interfaces • Production load and uptime still require formal benchmarking

Product Coverage

• Administrator and customer workflows • Desktop and web interfaces • Connected parking operations • Payments, notifications, and analytics

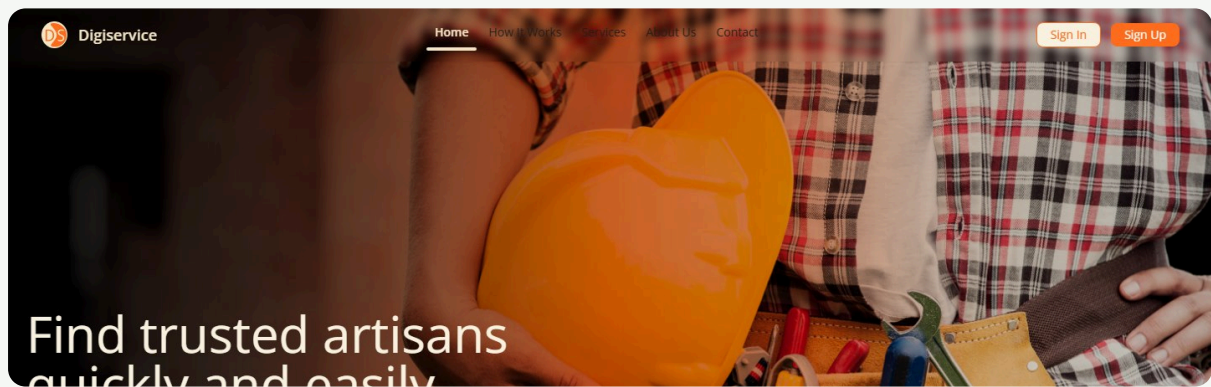
Demo [↗](#)

DigiService Reviews

Service Marketplace Microservice

DigiService is a team-built marketplace for connecting clients with local service professionals. I owned the reviews and evaluations service: a Java 17 Spring Boot API with review and response lifecycles, validation, filters, rating statistics, OpenAPI documentation, health endpoints, Eureka registration, and gateway-ready integration with other services.

CONTRIBUTION	Reviews microservice developer
DECLARED SCOPE	Owned backend service in a larger platform
PRIMARY CATEGORY	web



Representative portfolio screen

TECHNOLOGY EVIDENCE

Java 17

Spring Boot

Spring Data JPA

OpenFeign

Eureka

OpenAPI

Angular

IMPLEMENTED CAPABILITIES

Review Lifecycle

Create, read, update, delete, and filter reviews by artisan, client, status, or minimum rating.

Response Lifecycle

Separate endpoints manage replies to reviews by review or responding user.

Rating Statistics

Average rating and one-to-five-star distributions provide a reusable artisan reputation summary.

Input Validation

Rating ranges, required identifiers, response content, and request shapes are validated at the API boundary.

Cross-Service Checks

Feign clients validate linked users and artisans against adjacent platform services.

Discovery & Health

Eureka registration, gateway routing, Actuator health, and OpenAPI documentation support service operation.

ARCHITECTURE AND ENGINEERING

Owned Service in a Team Platform

The Angular marketplace calls a Spring Cloud gateway. Eureka routes requests to domain services, including the reviews service I built. That service owns review and response data while using Feign clients for cross-domain validation.

PROBLEMS SOLVED

Clear Service Ownership: Keeping reviews and responses independent while referencing users and artisans owned by other services.

Cross-Service Validation: Providing useful validation without duplicating user or artisan domain data inside the reviews service.

Discoverable API Contract: Documenting filters, statistics, response shapes, health, and error behavior for team integration.

OUTCOME AND CV-READY EVIDENCE

Complete Domain API

A focused reviews service covers CRUD, responses, filters, and reputation statistics.

Team-Ready Integration

Discovery, gateway routing, Feign boundaries, and OpenAPI documentation align the service with the wider platform.

Explicit Contribution

The case study separates my reviews-service ownership from the broader application built by the team.

SecondChance

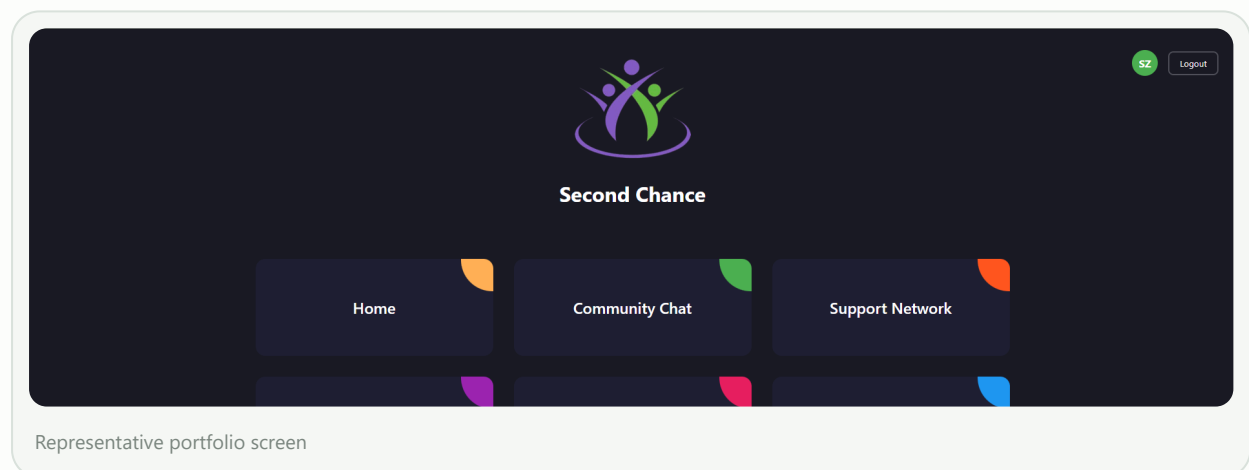
Reintegration Support Platform

SecondChance is a reintegration platform prototype designed around access to employment, training, support resources, and peer-community concepts. My work centered on FastAPI, MongoDB, server-rendered application flows, and the prototype user journey within the team project.

CONTRIBUTION Backend & application contributor

DECLARED SCOPE Social-impact platform prototype

PRIMARY CATEGORY web



Representative portfolio screen

TECHNOLOGY EVIDENCE

FastAPI

MongoDB

PyMongo

Motor

Jinja2

Pydantic

IMPLEMENTED CAPABILITIES

Job Opportunities

A prototype directory presents employment opportunities and the product direction for reentry-focused job access.

Community Space

A static community interface explores how peer-support conversations could fit into the wider platform.

Support Network

Prototype screens organize mental-health and legal-support resources in one discoverable directory.

Training Resources

Learning and skills content gives the prototype a practical path beyond job listings alone.

User Records

FastAPI and MongoDB routes cover registration, login lookup, and core user data operations for the demo.

Guided Dashboard

A unified dashboard connects jobs, training, support, community, and assistant interface concepts.

ARCHITECTURE AND ENGINEERING

Current Prototype Boundary

FastAPI, Jinja2, and MongoDB support the user and page flows, while jobs, resources, community, and assistant experiences are currently static interface prototypes. Production authentication, authorization, and realtime messaging remain future work.

PROBLEMS SOLVED

Sensitive Product Domain: Designing support flows that feel respectful, clear, and useful for people navigating reentry.

Prototype vs Production Security: The current login flow is demonstrative; password hashing, tokens, authorization, moderation, and privacy controls are required before real use.

Turning Static Concepts into Services: Jobs, training, support, and community screens define the product direction but still need data models and backend workflows.

ZenithHouse

Smart Home Automation System

ZenithHouse explores distributed home automation with ESP32 controllers assigned to room-level devices. A mobile application and Firebase provide user access, state synchronization, and remote control for climate, lighting, and door-access functions.

CONTRIBUTION **Embedded & mobile developer**

DECLARED SCOPE **IoT prototype**

PRIMARY CATEGORY **embedded + mobile**

TECHNOLOGY EVIDENCE

C++

Firebase

ESP32

IoT

Mobile App

IMPLEMENTED CAPABILITIES

Lighting Control

ESP32-connected lighting controls form one part of the smart-home project scope.

Climate Control

The project explores room-level climate control through connected devices.

Access Control

Door-access control extends the same device-to-application architecture.

Mobile Interface

A mobile application provides the user-facing control layer with Firebase state synchronization.

ARCHITECTURE AND ENGINEERING

Device-to-Mobile Concept

ESP32 firmware handles connected home controls while Firebase links device state to the mobile application. Detailed source material and new visuals are still being prepared for this archived project.

PROBLEMS SOLVED

Hardware–Application Boundary: Coordinating device commands, cloud state, and mobile feedback across the connected-home flow.

Case-Study Documentation: The CV verifies the project scope, while source files and updated screenshots still need to be added to this portfolio archive.

Smart Unlock

Keyless BLE Employee Access

Created during an internship at Scheidt & Bachmann, Smart Unlock uses a phone beacon simulator and an ESP32 WROOM-32 scanner to test a keyless access decision. A matched Eddystone identifier triggers an LED as the door-relay proof and publishes status through Mosquitto MQTT.

CONTRIBUTION Hardware & software developer

DECLARED SCOPE Connected access prototype

PRIMARY CATEGORY embedded

TECHNOLOGY EVIDENCE

C++

ESP32

Eddystone BLE

MQTT

Mosquitto

IMPLEMENTED CAPABILITIES

Beacon Scanning

ESP32 firmware scans nearby Eddystone advertisements and filters the expected beacon identifier.

Identifier Check

The proof of concept compares detected UID or URL data before allowing the hardware response.

Relay Simulation

An LED represents the door-relay action so the access decision can be tested safely on the bench.

MQTT Status

Mosquitto messaging publishes access events and prototype device status for remote observation.

Beacon Simulator

A phone-side beacon simulator provides repeatable BLE advertisements for testing the scanner flow.

ARCHITECTURE AND ENGINEERING

System Architecture

A phone beacon simulator broadcasts an Eddystone identifier. The ESP32 scans and filters the advertisement, triggers an LED as the lock-relay proof, and publishes event or status messages to Mosquitto over MQTT.

SPS

Smart Parking Mobile Application

Smart Parking Solution is a native Android project built with Java and XML. The documented prototype covers login, account creation, a parking-space view, logout, and vehicle records, with Firebase included in the application build.

CONTRIBUTION **Android & IoT developer**

DECLARED SCOPE **Android prototype**

PRIMARY CATEGORY **mobile + embedded**

TECHNOLOGY EVIDENCE

Java

Android

XML

Firebase

IMPLEMENTED CAPABILITIES

Login & Registration

Native Android screens support account entry and creation in the prototype journey.

Parking-Space View

A dedicated mobile view presents the current parking-space information available to the application.

Cars & Plates

Vehicle, plate, and owner records are organized into a mobile-friendly list.

Session Exit

The documented four-page journey includes a clear logout path.

ARCHITECTURE AND ENGINEERING

Android + Firebase Prototype

The native Java application covers account, parking, and vehicle-record views with Firebase included in the compiled prototype. Sensor ingestion, mapping, predictions, and alert services are outside the verified scope.

Smart Agri

Agricultural Monitoring Concept

The supplied Smart Agri folder is currently empty, so this archive entry intentionally avoids claiming a specific hardware, cloud, or mobile implementation until the original source and documentation are available.

CONTRIBUTION	Original implementation details pending
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DECLARED SCOPE	Source details pending
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PRIMARY CATEGORY	embedded + mobile
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IMPLEMENTED CAPABILITIES

Agricultural Monitoring

The archived project explored how a connected system could surface environmental field information to a user.

Documentation Pending

The local project folder does not currently contain source files or screenshots, so implementation details are intentionally not claimed.

ARCHITECTURE AND ENGINEERING

Archived Concept

This entry preserves the project in the portfolio archive without inventing hardware, cloud, power, or alert details that are not present in the supplied source folder.

SUPPORTING CONTEXT

Education and professional placements

PROFESSIONAL EXPERIENCE

Jul 2025 — Aug 2025

AI Automation & Full-Stack Developer

Inspark Connect — Tunis, Tunisia

Development of AI-powered workflows and intelligent assistants for the Prigado e-commerce platform using n8n, Laravel, Vue.js, MySQL, and Gemini.

Feb 2024 — May 2024

Internship — Embedded Systems & Software Developer

Scheidt & Bachmann — Megrine, Tunisia

Created MacroPark: a smart parking system using License Plate Recognition (LPR) and mobile app access with BLE fallback, MQTT communication, and ESP32-controlled barriers.

Jan 2023 — Feb 2023

Internship — Hardware & Software Developer

Scheidt & Bachmann — Megrine, Tunisia

Created Smart Unlock: a secure keyless door access system using BLE mobile authentication, MQTT communication, and ESP32 hardware control.

Jan 2022 — Feb 2022

Internship — Web Development

Sotunol — Megrine, Tunisia

Developed and improved the company's web platform, enhancing frontend interfaces and optimizing backend performance.

EDUCATION

Sep 2024 — Present

Engineering Degree in Web and Internet Technologies (TWIN)

ESPRIT — Ariana Governorate, Tunisia

Specializing in modern web architectures, distributed systems, cloud-native development, and advanced software engineering practices.

Sep 2021 — Jun 2024

Bachelor's Degree in Embedded and Mobile Systems

Higher Institute of Technological Studies of Radès — Radès, Tunisia

Focused on microcontroller programming, IoT architectures, mobile development, and hardware-software integration.

Sep 2017 — Jun 2021

Baccalaureate in Technical Sciences

Lycée Technique Farhat Hached Radès — Radès, Tunisia

Foundation in technical sciences, mathematics, and introductory engineering concepts.

Sofiene Zayati

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Prepared from the supplied project repositories and portfolio case-study data · July 2026